

Data is ART

- Friday Sept 4 @ 5pm, SDS
- Presentation and award ceremony

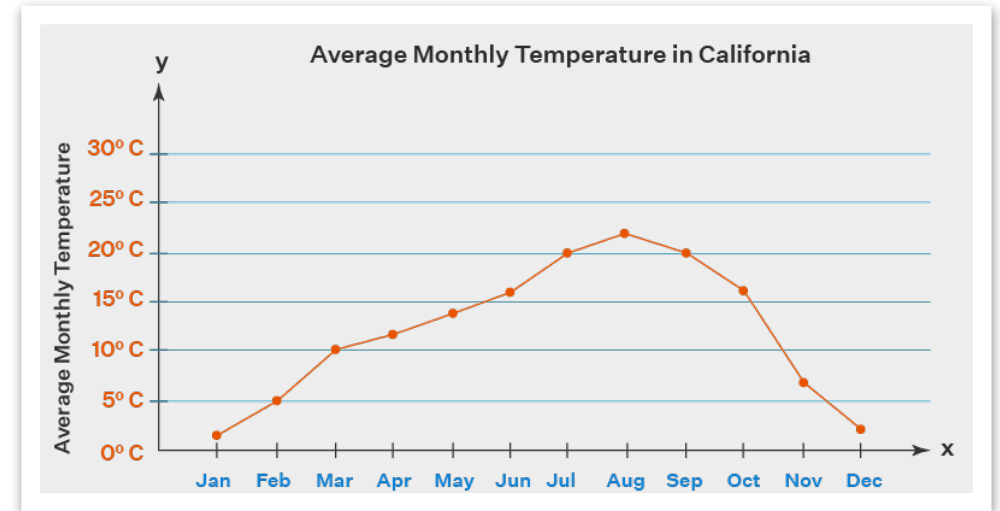


Two Kinds of Visualization

Some graphs are **direct transformations** of the data

These are designed to **describe and explore** data that has not be analyzed

We call these **data visualizations**
exploratory



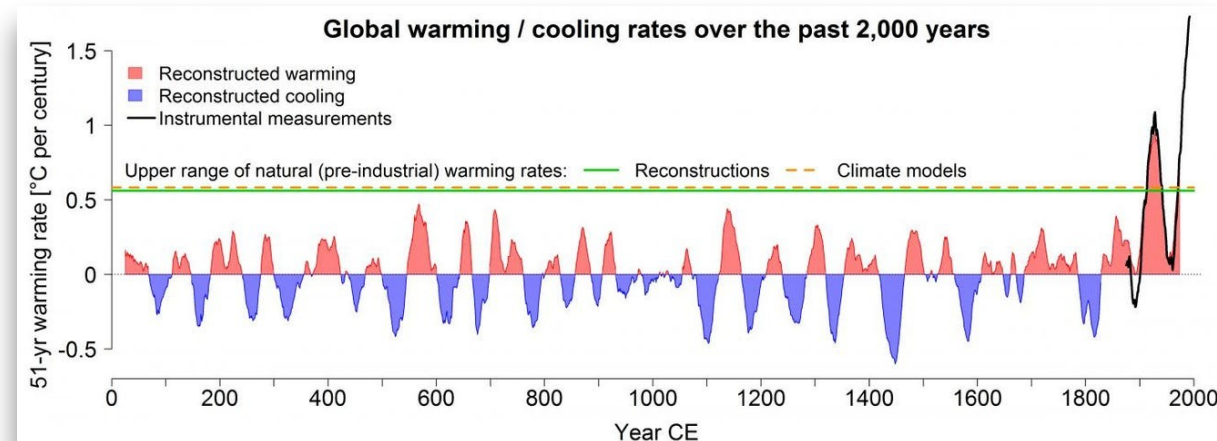
Others are highly **embellished data visualizations**

These are designed to **communicate and explain** results of analysis

These are examples of **storytelling with visualization**

We call these *explanatory*

Infographics, scientific figures, etc.



This **structure of this course** follows this pattern

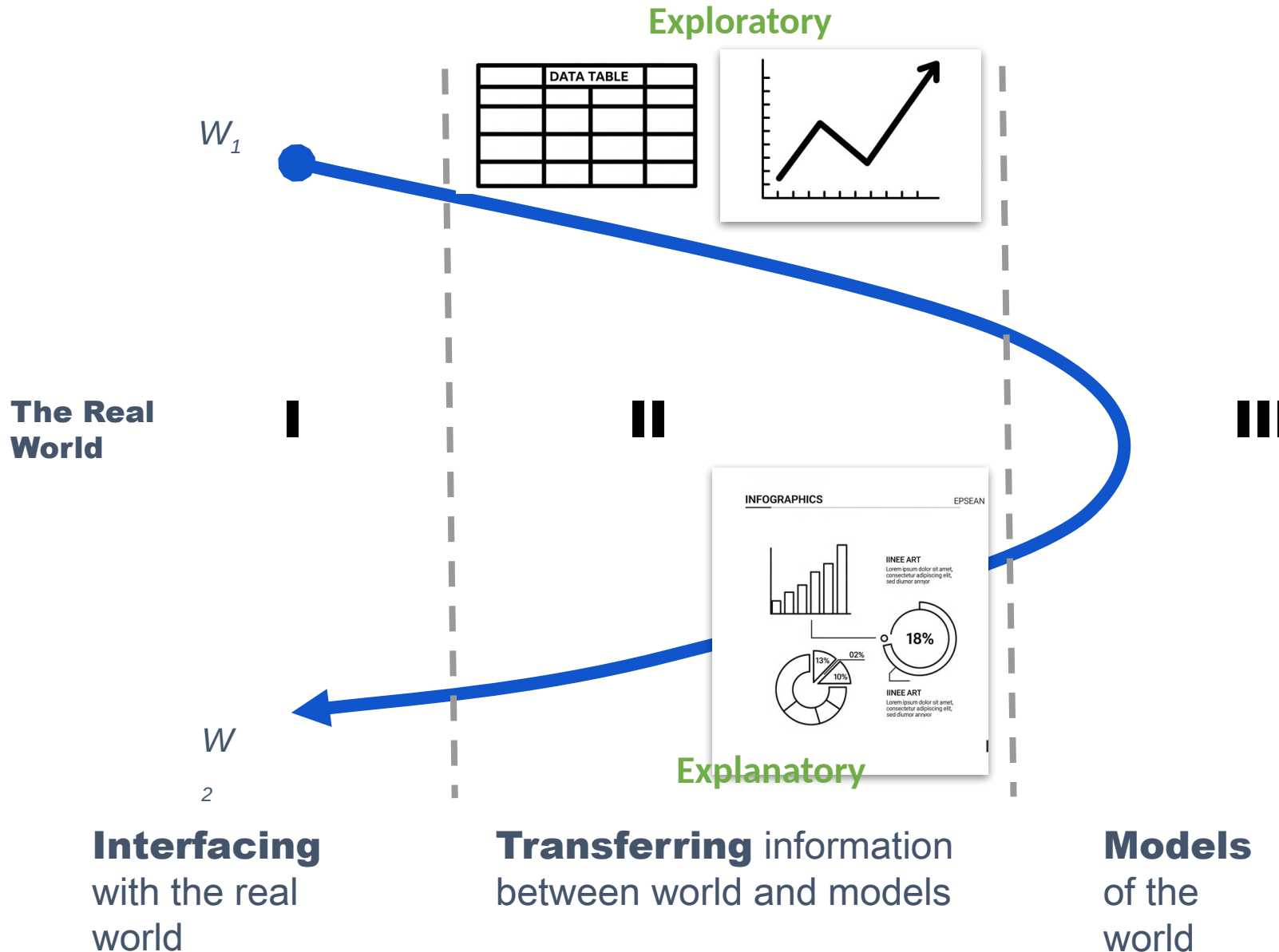
The **first half** will focus on **exploratory**
data visualization

The **second half** will focus on **explanatory**
storytelling and infographics

What is the *intent*?

Is the graphic displaying an underlying *model* of the data?

To explore data, we often need to code

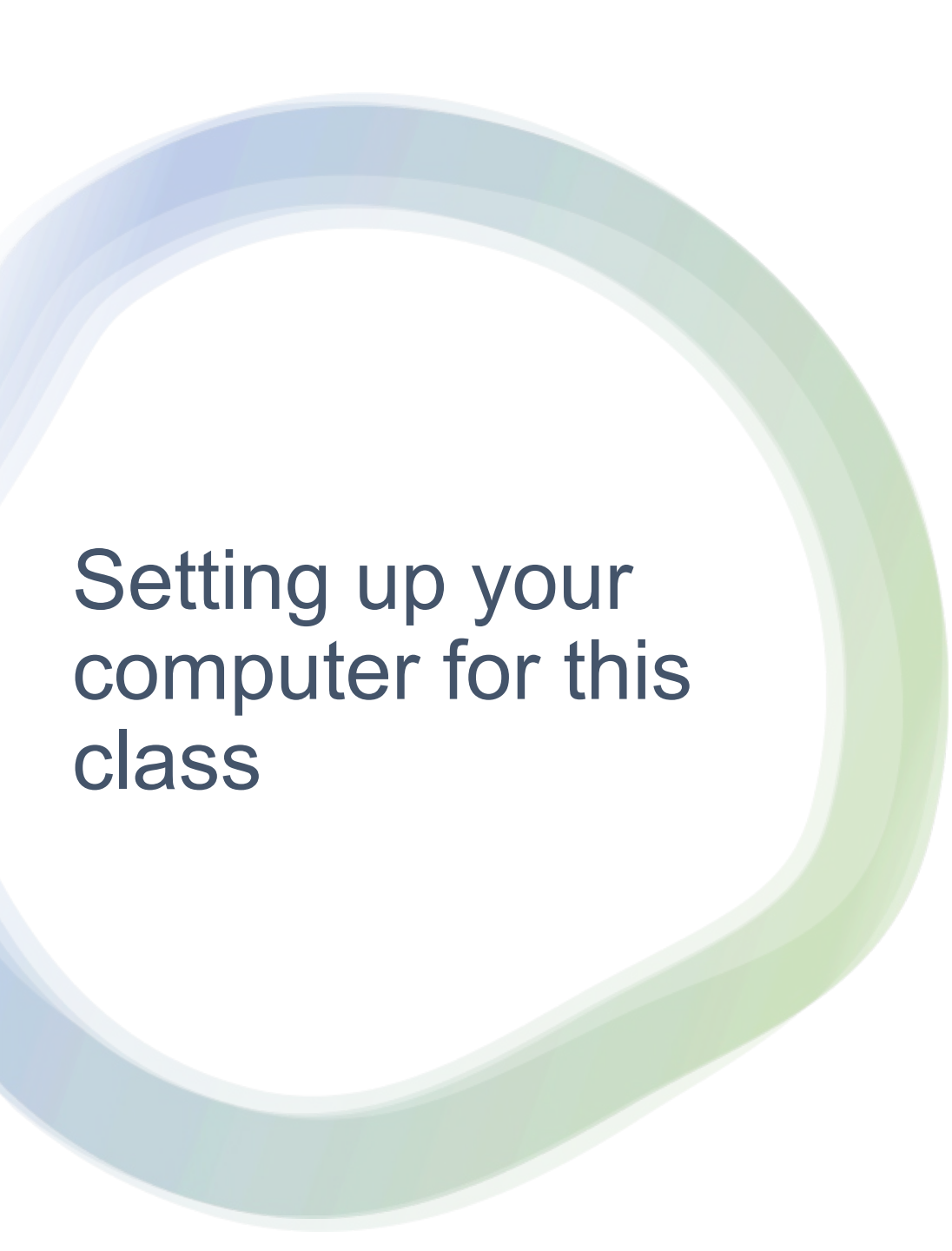


Files needed today:

`Lab_01.ipynb` [canvas]

Introduction to Coding in Python

Pre-Lab 1

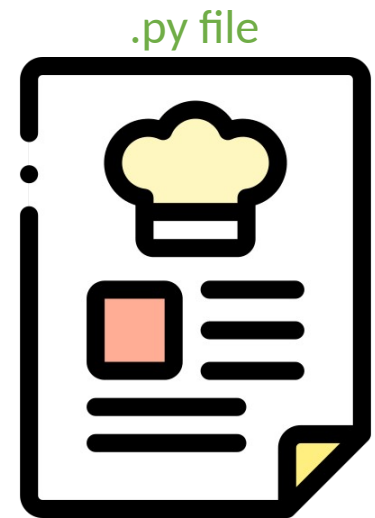


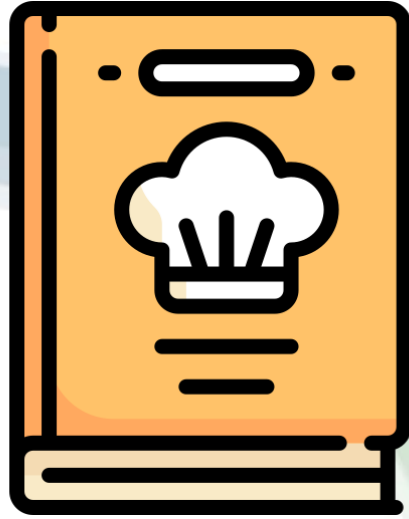
Setting up your computer for this class

- You should have:
 - **Anaconda** installed
 - **VS Code** installed (or an IDE you like)
- Today we will add:
 - A **folder** location to organize your files (code files and data files for this class)
 - VS Code extensions:
 - Jupyter notebook extensions
 - Optional extensions: CSV viewer
 - Python **packages** needed for this class

How do coding tools fit together?

- Writing code $\neq$ running code
- Think of coding like running a **restaurant**
 - .py files
 - Jupyter notebooks
 - VS Code (IDE)
 - Python interpreter
 - Terminal/Command Prompt
 - Anaconda & environments
 - GitHub and git
- Your **code** = the recipe
 - It describes ingredients and steps, but a recipe does not cook itself!





Folder

Organization: The recipe book

- Pick **one location** to make a folder for this class (Documents? UVA Box? Home?)
 - Each lab will have its own subfolder that we will put in this folder
 - Any extra data/files from Canvas should go here
- Think about what you want to name your files!
- You will open this folder every time you work on this class!



Python interpreter: The Chef

- The chef reads the recipe line by line and executes each instruction
- Different chefs exist:
 - Python 3.7
 - Python 3.9
 - Python 3.11 ...
- Same recipe with different chef → slightly different outcomes!



Terminal & Command Prompt: The Order Counter

- The terminal is where orders are placed

```
python analysis.py
```

means “when you run, make the `analysis.py` recipe”

- We can access the terminal through stand-alone apps (terminal, command prompt) or *within VS code*
 - Like different windows at the same order counter

The Restaurant: The Big Picture

You write recipes
(code)...



... and put them
in a recipe book
(folder)...



... which you can send
to the order counter
(terminal)...



...which tells the chef
(Python) what to
cook

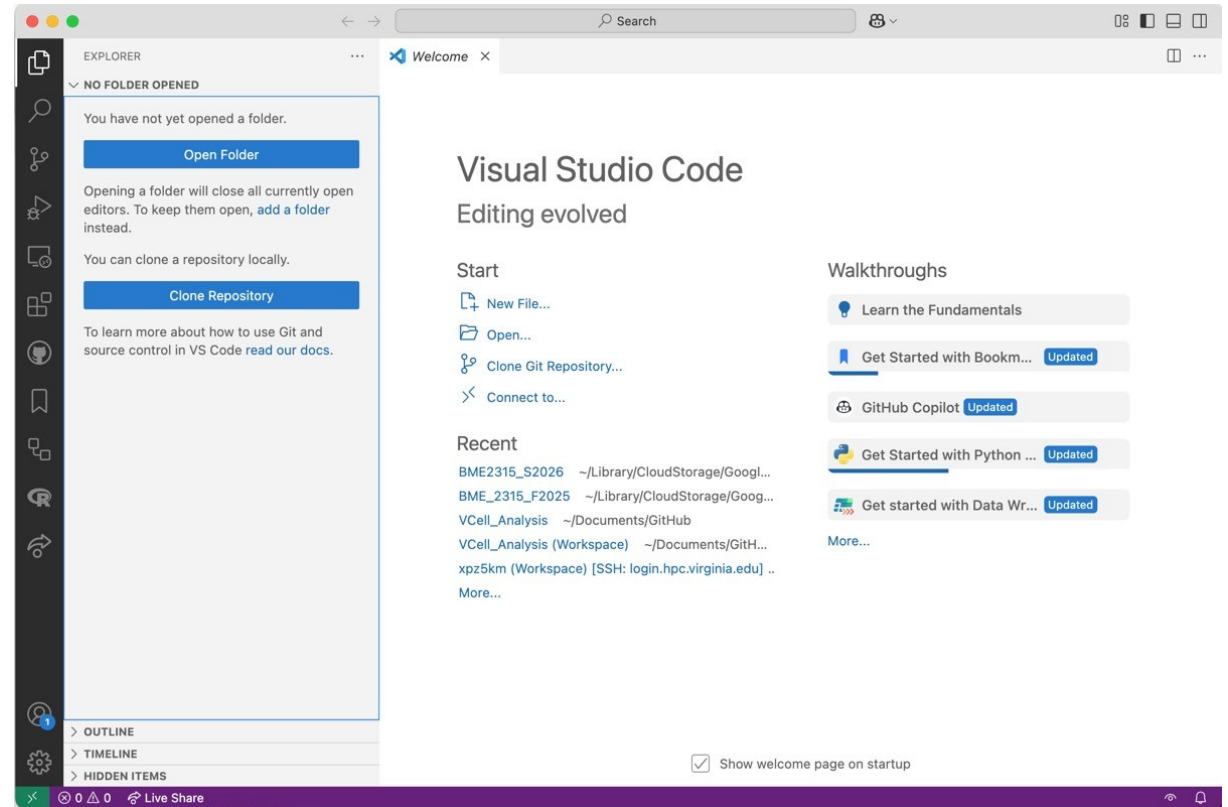


The Restaurant (VS Code)

All of this happens inside the restaurant, VS Code!

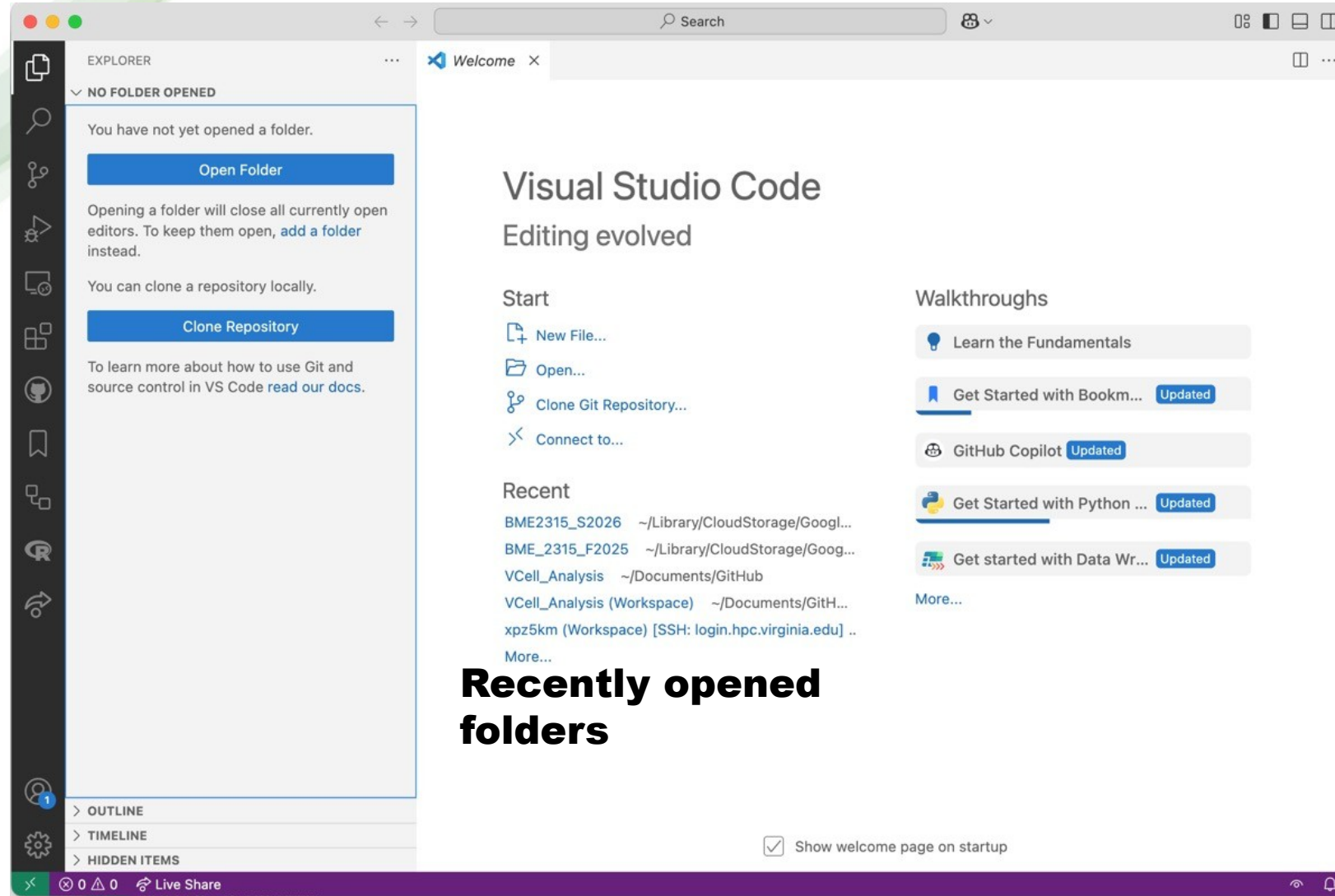
VS Code & Integrated Development Environments: The Restaurant

- VS code is a place to work on code (write recipes), view data files (peek in the kitchen), and run your code through a terminal (tell the chef what to cook)
- **Integrated**: many tools in one place
- **Development**: place to build code
- **Environment**: workspace



VS Code IDE

Welcome



Recently opened folders

Always start by opening your class folder from VS code!

Let's practice!

Make a **folder** for this class on your computer (not in Downloads please!!)

Open **VS Code** (or another editor).

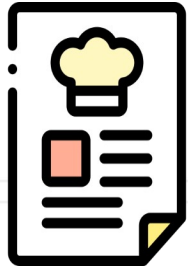
If you have a PC, open VS code THROUGH **ANACONDA NAVIGATOR**.

Stand your name card up on its edge when you are done. Raise your hand for help!

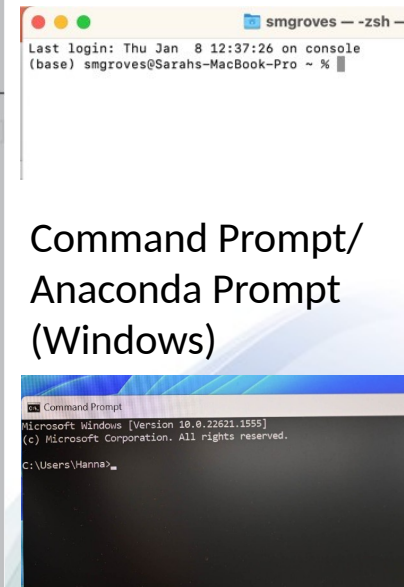
VS Code IDE

VS Code IDE interface with annotations:

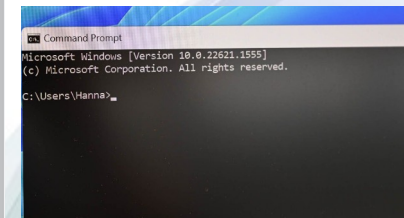
- File Explorer**: Left sidebar showing the project structure. Annotations include "Your chosen folder!" and "Add a file", "Add a folder", "Refresh", and "Collapse all folders".
- Search**: Search bar at the top of the File Explorer.
- Extensions**: Extensions sidebar at the bottom left.
- Script Editor**: Central area for editing code. The file "addTwoNumbers.py" is open, showing a Python function. Annotations include "Search or run a command" and "Name of file".
- Run File**: Button to execute the current file.
- Console/Terminal/Output**: Bottom panel showing the terminal output of the executed script.
- Python version**: Status bar at the bottom right showing the active Python interpreter.



Terminal (Mac)



Command Prompt/ Anaconda Prompt (Windows)



Let's practice!

Open the **folder** for this class in VS code.
Add **lab0_blank_python.ipynb** from canvas into this folder and **open** it in VS code.
Go to the **extensions** marketplace and download **Python & Jupyter**.
Stand your name card up on its edge when you are done. Raise your hand for help!

A positive feedback loop!

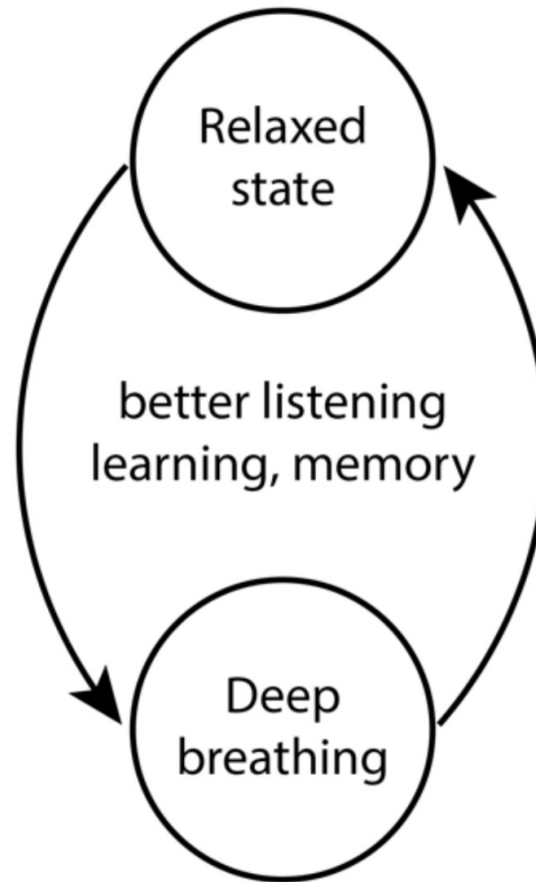


Figure 1. The relaxed state is better for listening, learning and remembering.

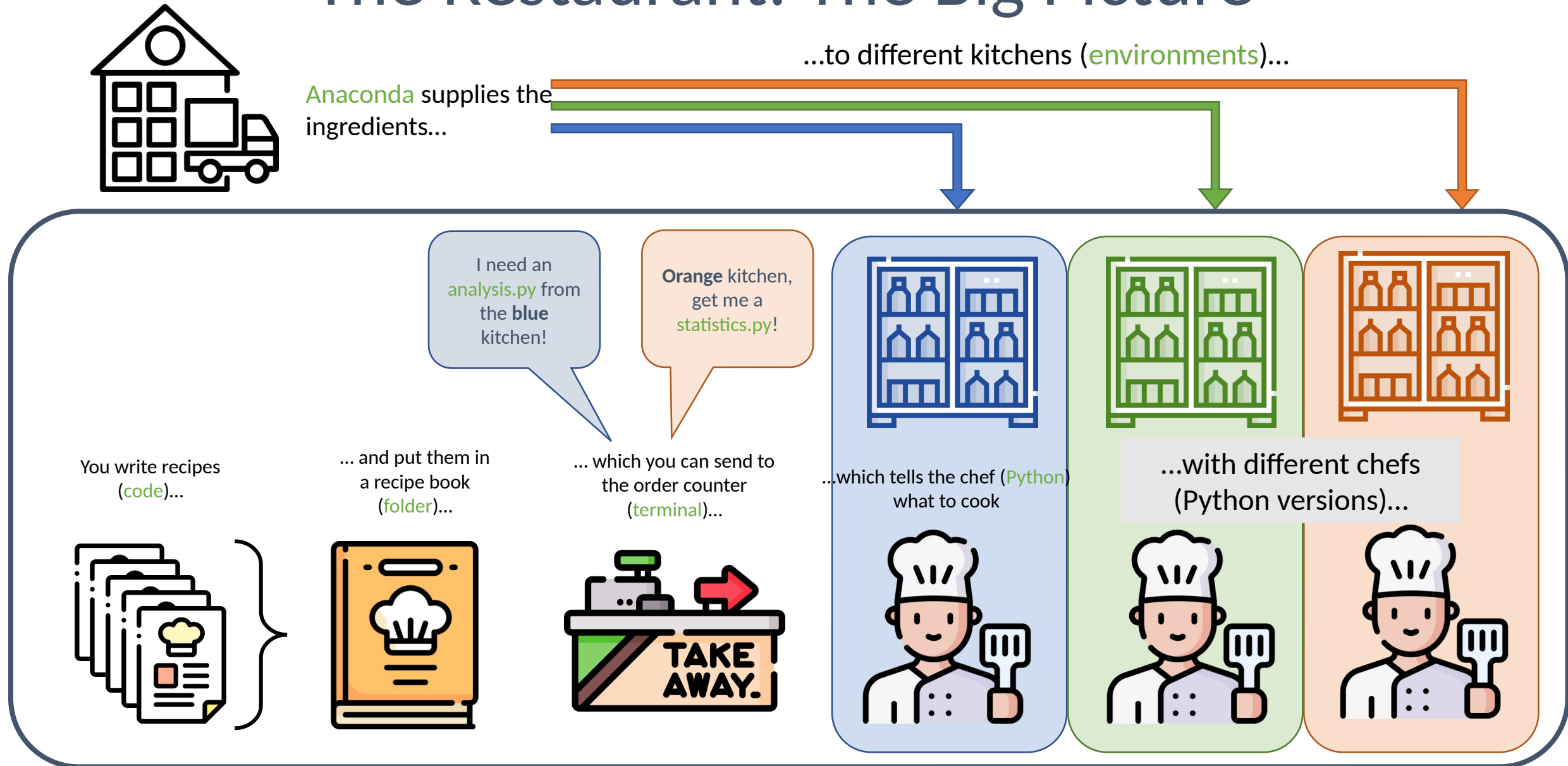


Anaconda: The supplier & pantry Conda environments: The kitchens

- Anaconda is the supplier that includes Python and scientific libraries
 - The package manager gives you a stocked pantry
- Conda environments = separate kitchens
 - The *base* environment is the default
 - You can make as many environments as you want. This is useful for organizing projects.
 - Each kitchen can have different chefs (Python versions) and different pantries
 - Ensures there are no “ingredient conflicts”
- VS code works in *one kitchen at a time*



The Restaurant: The Big Picture



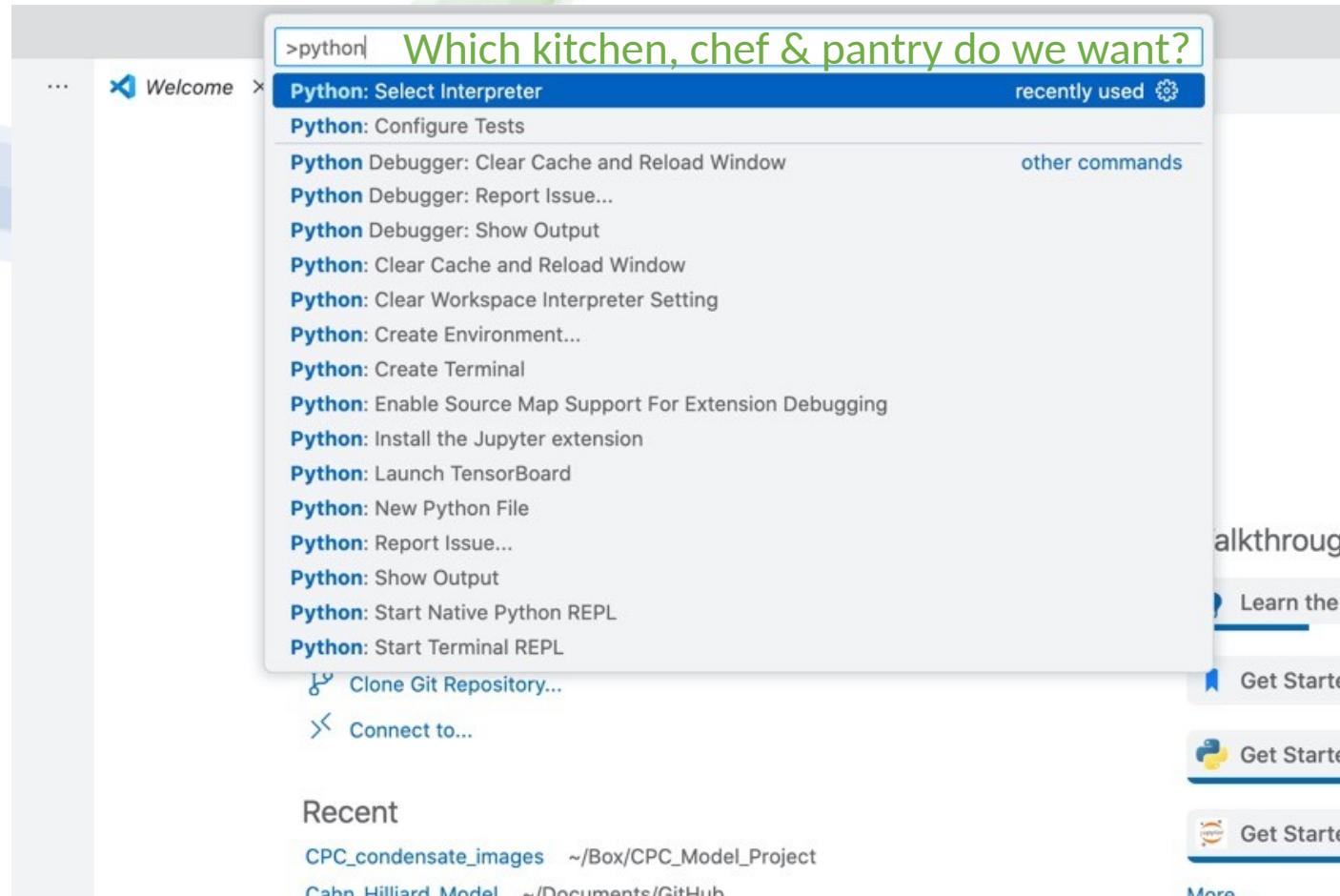


Importing packages (adding food to the pantry)

- *conda* and *pip* are two major “warehouses”
 - Most python packages are hosted on Python Package Index ([pypi](https://pypi.org/)) and can be installed with *pip*
 - Ensure the right conda environment is activated to add the package in the right location
 - If a package is not available through pip, try *conda* (or look at the install guide)

```
Last login: Thu Jan  8 13:59:41 on ttys002
(base) smgroves@Sarahs-MacBook-Pro ~ % conda activate BME2315
(BME2315) smgroves@Sarahs-MacBook-Pro ~ % pip install seaborn
Requirement already satisfied: seaborn in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from seaborn) (1.24.0)
Requirement already satisfied: pandas>=1.2 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from seaborn) (1.5.3)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from seaborn) (3.7.1)
Requirement already satisfied: contourpy>=1.0.1 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from seaborn) (1.1.0)
Requirement already satisfied: cycycler>=0.10 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from seaborn) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from matplotlib!=3.6.1,>=3.4) (4.22.0)
Requirement already satisfied: packaging>=20.0 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from contourpy>=1.0.1) (23.1)
Requirement already satisfied: pillow>=8 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from matplotlib!=3.6.1,>=3.4) (9.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from matplotlib!=3.6.1,>=3.4) (3.1.0)
Requirement already satisfied: python-dateutil>=2.7 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from pandas>=1.2) (2.8.2)
Requirement already satisfied: importlib-resources>=3.2.0 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from pandas>=1.2) (5.12.0)
Requirement already satisfied: pytz>=2020.1 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from pandas>=1.2) (2023.3)
Requirement already satisfied: tzdata>=2022.7 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from pandas>=1.2) (2023.3)
Requirement already satisfied: zipp>=3.1.0 in ./Documents/anaconda3/envs/BME2315/lib/python3.9/site-packages (from importlib-resources>=3.2.0) (3.15.0)
(BME2315) smgroves@Sarahs-MacBook-Pro ~ %
```

VS Code IDE

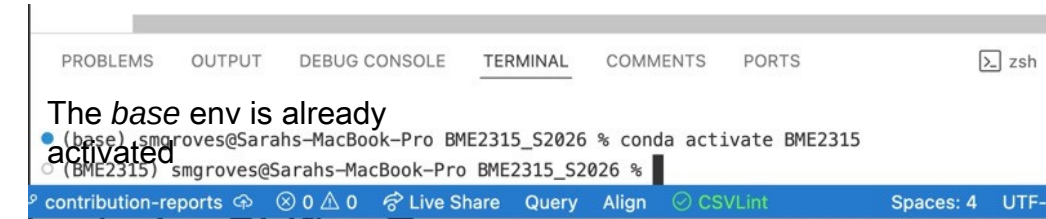


Command palette: Command/Control + Shift + P

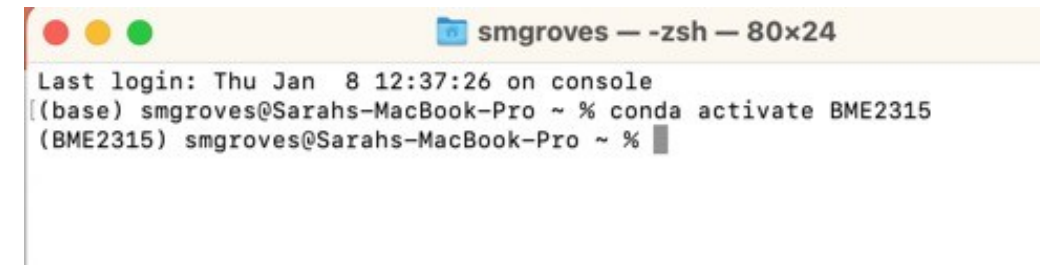
Configures the right version of Python for the entire IDE

Do this once!

Terminal



I am activating a special env (BME2315)



“conda activate DS2003”

“source activate DS2003”

Configures the right version of Python for
this **specific** terminal session

Do this when installing packages

Let's practice!

1. Using the **terminal** (in VS code, anaconda prompt, or terminal), make a conda environment and activate it:

```
conda create --name PAX6_grn_env python=3.12
```

```
conda activate PAX6_grn_env
```

2. Add important packages from pip:

```
pip install pandas numpy scipy ipykernel matplotlib seaborn plotly
```

3. Using the **command palette in VS code** (Command/Control + Shift + P), select the base environment in VS code:

```
Python: Select Interpreter > choose (DS2003)
```

4. Open the Lab, click on **Select Kernel** in the top right corner, and pick (Python environments > DS2003)

Stand your name card up on its edge when you are done. Raise your hand for help!

A positive feedback loop!

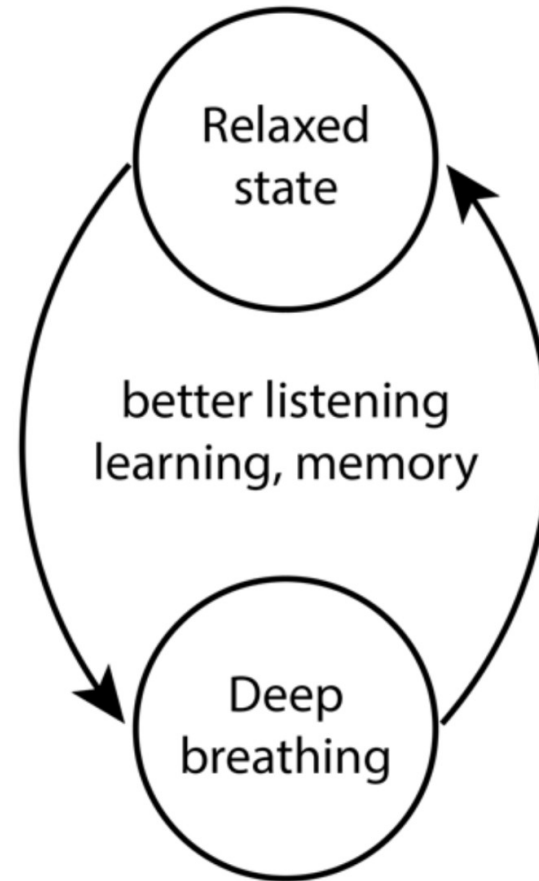
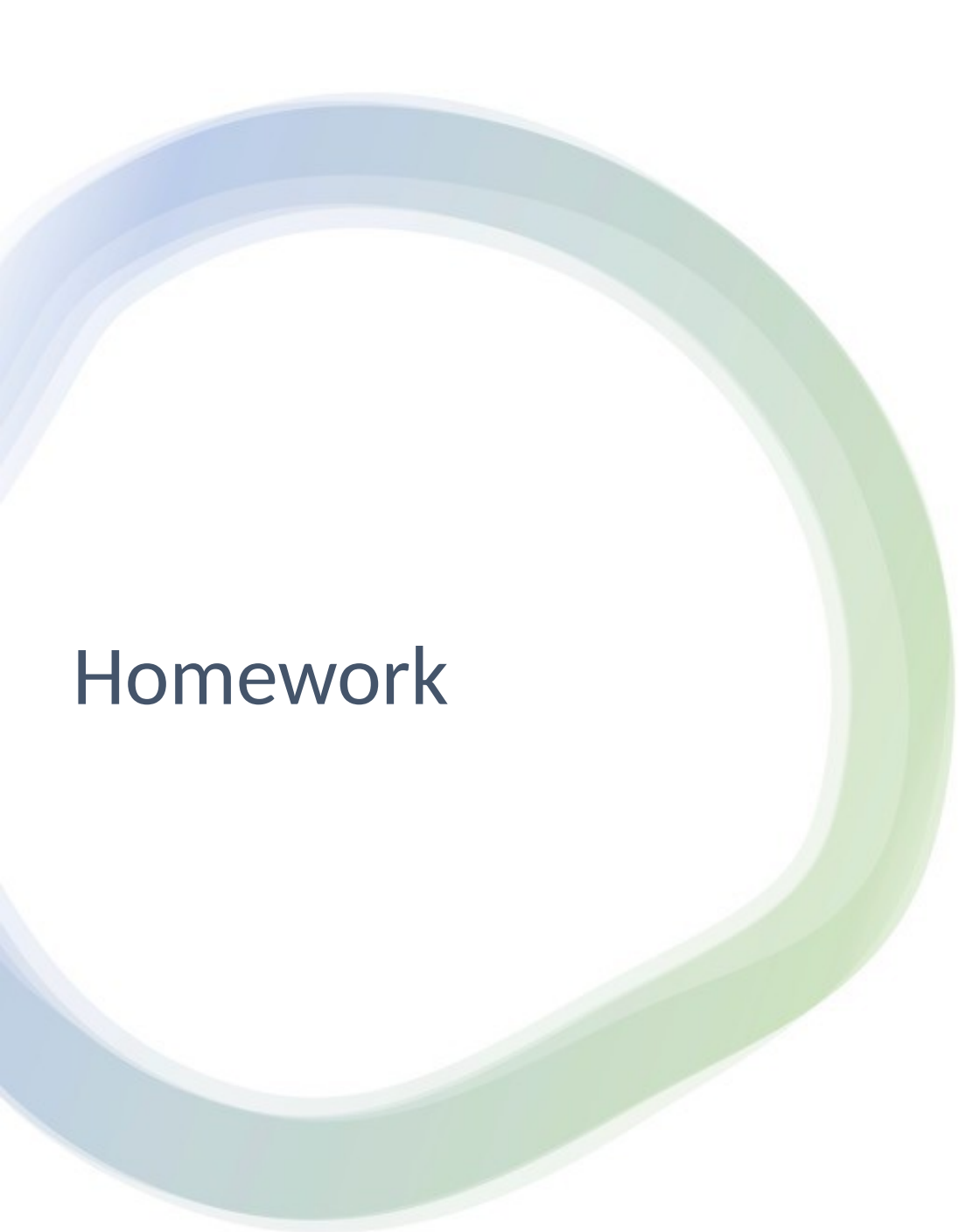


Figure 1. The relaxed state is better for listening, learning and remembering.

- I fully expect problems with computers
- Everyone's computer is set up in a different way
- If you did not get your setup done, please see me after class or email me!
- Most likely you did not mess something up and I can help you figure out what went wrong



Homework

- You may work with others, but please add all of their names
- You may use genAI, but you must *cite* it
 - You are also accountable for all code and writing in the labs
- This lab is due by **Sunday at midnight**